

Removing Layers to Find the Workspace — Llama-3.1 Layer-Ablation Study

Removing Layers to Find the Workspace: A Layer-Ablation Study of Llama-3.1 (8B / 70B / 405B)

Abstract

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Removing Layers to Find the Workspace: A Layer-Ablation Study of Llama-3.1 (8B / 70B / 405B)

A self-contained research report. We locate where meaning forms across the depth of Llama-3.1, delete layers — singly and in bands — and measure the effect on the next-token distribution and on free generation, across three model sizes and two compute backends (a local T4 GPU and NDIF remote via nnsight) . We connect the results to Anthropic’s 2026 “global workspace” (J-space) finding.

Author: Sumit Yadav · **Model family:** meta-llama/Llama-3.1 · **Tools:** PyTorch, Transformers, nnsight/NDIF · **Compute:** 1× NVIDIA T4 (4-bit) + NDIF remote (fp16, up to 405B).

Abstract

Transformer language models refine a *residual stream* layer by layer. We ask three questions: (1) *Where* along the depth does class/semantic structure become available? (2) *Which* layers are causally necessary — what happens if we remove them? (3) Does the answer change with model *scale*? Using a layer-wise projection of hidden states we show that topic information is linearly **decodable from layer 2 onward yet visually entangled** through the first half of the network, re-separating only near the output. Ablating a **single** middle layer is nearly harmless (an “early cliff”: only the first two layers are individually critical), but removing a **contiguous middle band** degrades the model monotonically. Splitting by task type, **abstract / creative generation degrades first and single-step recall last** — the qualitative signature Anthropic report for their J-space “global workspace.” Replicating on full-precision 8B, 70B and 405B via NDIF shows the pattern is not a quantization or instruction-tuning artifact, and that **larger models tolerate more middle-band removal** before their next-token distribution collapses — while *free-generation fluency* is far more robust than the distributional metrics imply.

1. Objective

1. **Localise representation formation.** Determine, per layer, where a 4-way topic label is
 1. *decodable* and (b) *visually separable*, and interpret the gap between the two.
 2. **Measure causal necessity by ablation.** Quantify the effect of making one layer — or a band of contiguous layers — an identity function, on (i) the next-token distribution (KL divergence, top-1 agreement, entropy, perplexity) and (ii) free generation (readable text).
 3. **Test capability-specificity.** Ask whether removing the middle band harms *reasoning* differently from *recall*, *math*, *translation*, and *creative* generation.
 4. **Test scale-dependence.** Repeat the ablation on 8B, 70B and 405B and characterise how tolerance to removal grows with model size.
 5. **Relate to the “global workspace” hypothesis** (Anthropic 2026) and to the layer-pruning literature.
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2. Background and Literature Review

The residual stream. A decoder block ℓ reads the running hidden state and writes an additive update: $x_\ell = x_{\ell-1} + \text{Attn}(x_{\ell-1}) + \text{MLP}(x_{\ell-1})$. Because the update is additive, each block can be *removed* by replacing its output with its input ($x_\ell = x_{\ell-1}$) — the stream simply skips it. This “residual stream as a communication bus” view underlies most of modern interpretability (Elhage et al., *A Mathematical Framework for Transformer Circuits*, 2021).

Reading intermediate states. The **logit lens** (nostalgebraist, 2020) decodes an intermediate residual through the output unembedding to see the model’s “current guess”; the **tuned lens** (Belrose et al., 2023) learns per-layer probes to do this faithfully. Tenney, Das & Pavlick (2019) showed BERT “rediscovers the classical NLP pipeline” across depth — lower layers do surface/syntax, higher layers do semantics — motivating the idea that different depths carry different kinds of information.

Superposition and features. Elhage et al. (*Toy Models of Superposition*, 2022) argue features are packed into overlapping directions, so information can be present (decodable) even when a 2-D projection looks entangled. This directly predicts the decodability-vs-separation gap we observe in §5.1.

Causal methods. ROME (Meng et al., 2022) uses **activation patching** / **causal tracing** — corrupt an input, restore a clean activation at a specific (layer, position), and see whether the output recovers — to localise where a fact “lives.” Ablation (setting a component to identity / zero / mean) is the complementary “necessity” test.

Layer redundancy / pruning. Gromov et al. (*The Unreasonable Ineffectiveness of the Deeper Layers*, 2024) and Men et al. (*ShortGPT*, 2024) show many transformer layers — especially deeper/middle ones — can be pruned with little loss, and that larger models are more redundant. Our single-layer and scale results are consistent with this.

The global workspace. Anthropic’s 2026 report *Verbalizable Representations Form a Global Workspace in Language Models* (Gurnee, Sofroniew, ... Lindsey) identifies a **J-space** — a small (~6–10% of variance) set of token-aligned directions in intermediate layers, discovered with a **Jacobian lens** — that behaves like a cognitive-science *global workspace* (Baars 1988; Dehaene): a shared blackboard the model reads concepts from and reasons over. Suppressing the top J-space directions **impairs multi-hop reasoning and abstract generation while preserving parsing, fluency, and single-step recall**. This report is a coarse, layer-level analogue of that surgical, direction-level experiment; the repository name (...-remove-j-space-layer) reflects the connection.

3. Methodology and Mathematics

3.1 Layer indexing

With `output_hidden_states=True` a model with L decoder layers returns $L+1$ tensors: index 0 is the token-embedding output, and index k ($1..L$) is the output of decoder layer $k-1$. We write **L k** for the k -th hidden-state tensor (plot label) and use the **0-based decoder index** $\ell \in [0, L-1]$ for ablation. Thus plot **L15** = **output of decoder layer** $\ell=14$. (This off-by-one is a common source of error.)

3.2 The ablation operator (Figure 1)

To remove decoder layer ℓ we make it the identity map on the residual stream. Two equivalent implementations were used:

- **Discard-delta wrapper (local, PyTorch)**. Wrap the block so it still *runs* (keeping the attention KV-cache consistent during generation) but returns its **input** unchanged: `output_hidden_states := input_hidden_states`, i.e. $x_\ell = x_{\ell-1}$. Fully reversible.
- **Rebind (remote, nnsight)**. Since layer ℓ 's input equals layer $\ell-1$'s output, set `layers[ $\ell$ ].output = layers[ $\ell-1$ ].output` (embeddings for $\ell=0$). Identical net effect.

To remove a **band** $\{\ell_{\text{start}} \dots \ell_{\text{end}}\}$ we set every block in the band to pass the pre-band value through, so the whole band collapses to identity.

3.3 Distributional metrics

Let p and q be next-token probability distributions over the vocabulary V .

- **Entropy** (uncertainty of one distribution):
 $H(p) = - \sum_{x \in V} p(x) \log p(x)$.
- **Cross-entropy**: $H(p, q) = - \sum_x p(x) \log q(x)$; the model's **perplexity** on data is $\text{PPL} = \exp(H)$.
- **KL divergence** ("relative entropy," how far p is from reference q):
 $D_{\text{KL}}(p \parallel q) = \sum_x p(x) \log(p(x) / q(x)) = H(p, q) - H(p)$. It is ≥ 0 , zero iff $p = q$, and **not symmetric**. *KL is not entropy* — entropy is about one distribution, KL about two.

We compare an **ablated** model to the **intact baseline**, teacher-forced on the baseline's own greedy continuation (same context, so differences are purely the ablation's effect):

- **Mean KL** = $\text{mean}_t D_{\text{KL}}(p^{\{\text{abl}\}}_t \parallel p^{\{\text{base}\}}_t)$ over continuation positions t — how much the predictions moved.

- **Top-1 agreement** = $\text{mean}_t \mathbb{1}[\arg\max p^{\{\text{abl}\}}_t = \arg\max p^{\{\text{base}\}}_t]$ — fraction of positions whose most-likely token is unchanged.
- **Entropy** $H(p^{\{\text{abl}\}})$ — whether the broken model becomes *uncertain* (entropy $\uparrow$) or *stuck* (entropy $\downarrow$, repetition). KL catches both; entropy says which.

3.4 Representation-geometry metrics

For a labelled set (4 topic classes), per layer we compute:

- **Decodability** — 5-fold cross-validated **k-NN accuracy (k=15)** on the *full* hidden dimension (standardised). High accuracy $\Rightarrow$ the class is linearly-ish readable there.
- **Visual separation** — the **silhouette score** on the 2-D projection: for a point i , $s(i) = (b(i) - a(i)) / \max(a(i), b(i))$, where a is mean intra-cluster distance and b the mean distance to the nearest other cluster; we report the mean over points (range $[-1, 1]$; ~ 0 means clusters overlap).
- **Projection** — **UMAP** (McInnes et al., 2018), a neighbour-graph method, after PCA-50; a nonlinear 2-D embedding for visualization only (t-SNE gives a similar picture).

3.5 Interventions used in the appendix study (steering, patching, attribution)

- **Steering**: add a concept direction d (e.g. mean activation difference between positive and negative exemplars, unit-normalised) to the residual: $x_\ell \leftarrow x_\ell + \alpha \cdot d$.
- **Activation patching**: copy a clean-run activation into a corrupted run at (layer, position) and measure recovery of the target probability — localises causal information.
- **Attribution by ablation**: ablate each layer and record the change in the target token’s probability — a necessity-based importance score.

4. Experimental Setup

- **Models**. Llama-3.1-8B-Instruct (local, 4-bit nf4 on a T4) for the interactive/ablation work; base Meta-Llama-3.1-8B / 70B / 405B on NDIF remote (fp16) for replication and scale.
- **Compute**. A single 16 GB NVIDIA T4 (hence 4-bit for the 8B locally); NDIF remote execution via nnsight for full precision and for 70B/405B (no local GPU can hold those).
- **Representation dataset**. 400 texts, 4 balanced classes from 20-Newsgroups (*space, baseball, graphics, mideast*), mean-pooled per layer.

- **Ablation eval sets.** A small fixed prompt set (8) and a larger **100-prompt** set balanced over five capabilities (recall / reasoning / math / translation / creative, 20 each). Greedy decoding throughout for reproducibility.
- **Serving.** A FastAPI server exposes /generate returning baseline + ablated text and the metrics; a single-file browser UI drives it over an SSH tunnel; a multi-layer toggle grid selects which decoder layers to remove.

5. Experiments and Results

Figure 1 — The method (residual stream and the ablation operator)

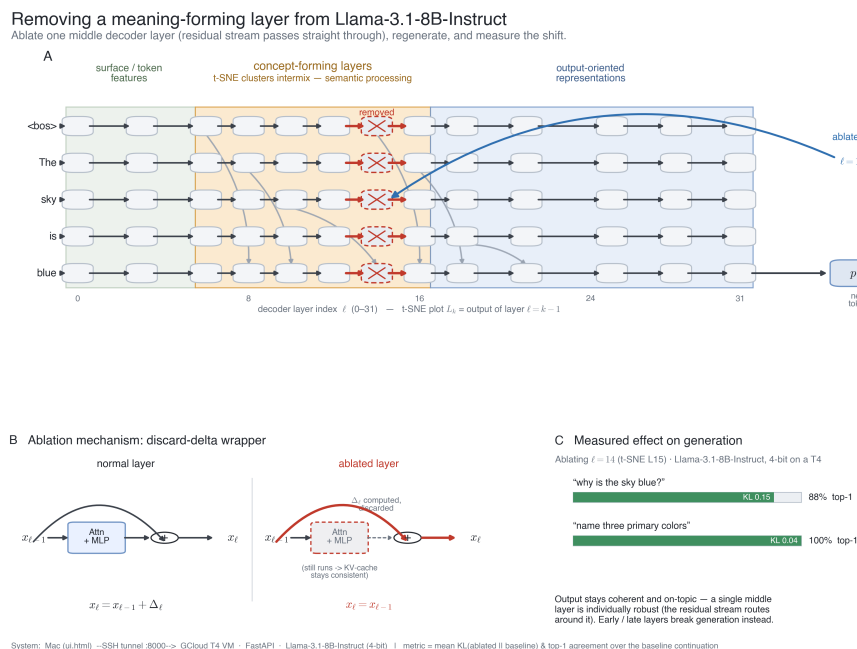


Figure 1

Description. Left/top: the token $\times$ layer grid of the residual stream, shaded into three depth regimes (surface L0–L5, concept-forming L6–L17, output-oriented L18–L32), with attention arrows flowing into the final (generation) position and the ablated layer drawn as a crimson dashed “skipped” column whose residual passes straight through. Bottom-left: the discard-delta mechanism — a normal block computes $x_\ell = x_{\ell-1} + \Delta_\ell$, the ablated block returns $x_\ell = x_{\ell-1}$ (Δ discarded, but the block still runs so the KV-cache stays consistent). Bottom-right: the measured effect readout. This figure defines every operation used below.

5.1 Where does meaning form? (Figures 2–3)

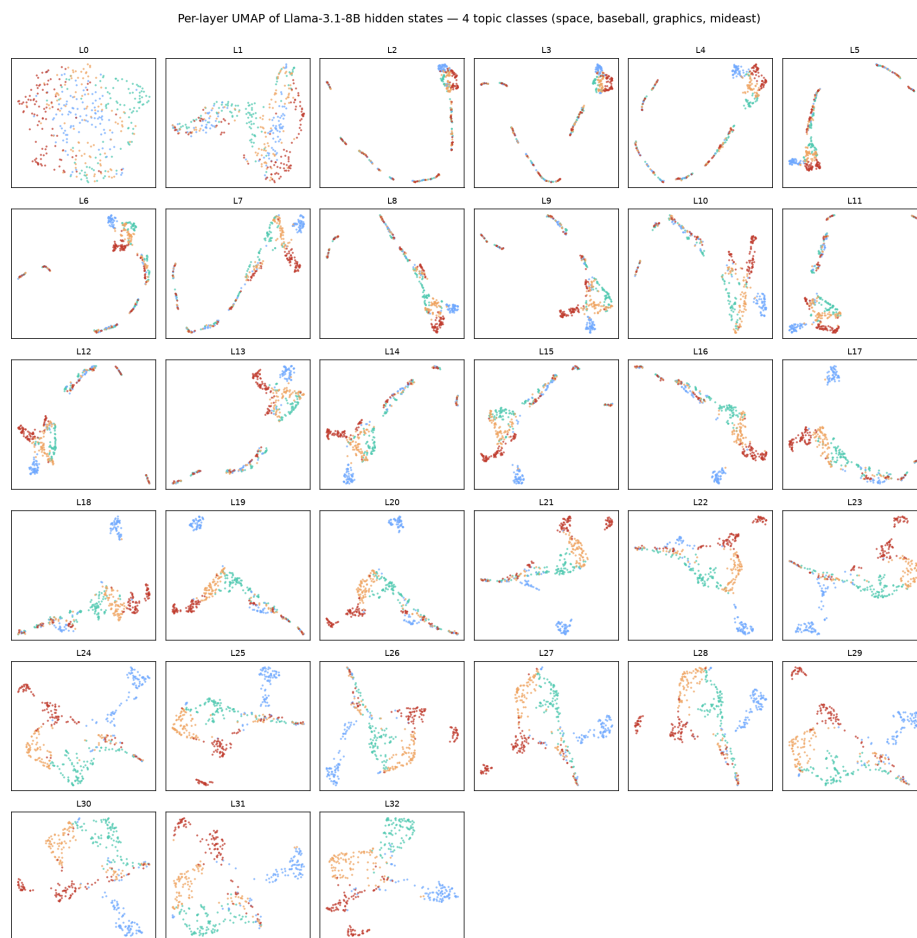


Figure 2

Figure 2 — per-layer UMAP of hidden states, one panel per layer, coloured by topic class. The embedding output (L0) is a diffuse cloud; by L2–L6 the points collapse onto thin, folded one-dimensional manifolds with the four topics **interleaved**; through the middle they remain entangled; and from ~L16 to the final layer they resolve into **clean four-way clusters**. The visual arc is *entangle* → *re-specialise*.

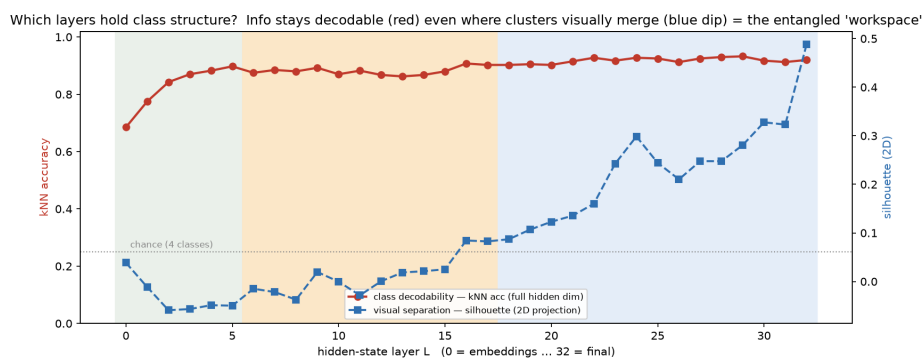


Figure 3

Figure 3 — *decodability vs visual separation across layers*. Red (left axis): k-NN accuracy on the full 4096-d hidden state rises from **0.68 at L0** to **~0.90 by L5** and stays **0.86–0.93** for the rest of the network — the class is *always readable* from layer 2 on. Blue (right axis): silhouette on the 2-D projection is **~ 0 (even slightly negative) through ~L1–L15**, then climbs steeply to **0.49** at the final layer. The **gap** between the curves is the key result: information is present but **distributed/entangled** in the first half (superposition), becoming geometrically separated only near the output. This is the layer-level shadow of the low-variance, information-rich “workspace.”

5.2 Removing one layer at a time (Figures 4, 6)

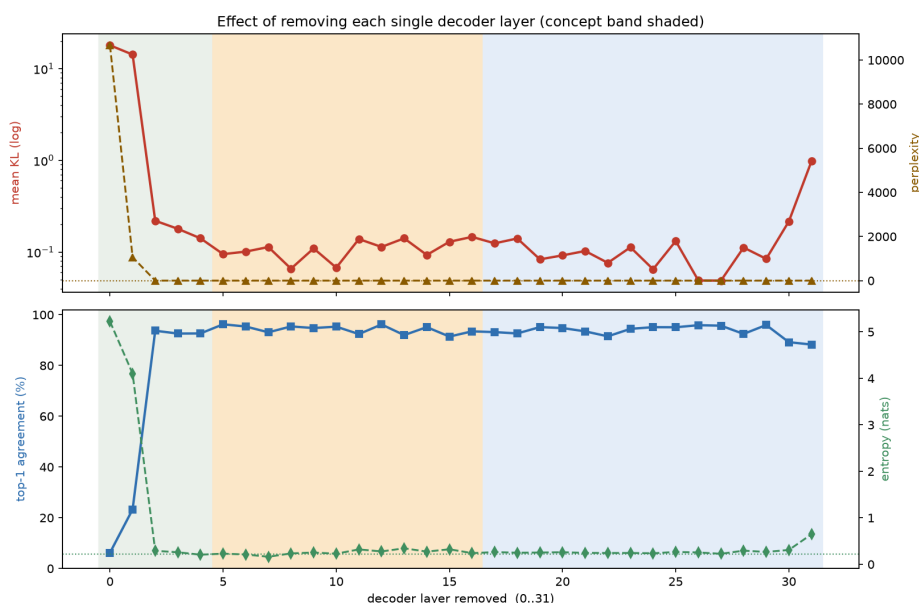


Figure 4

Figure 4 — *single-layer ablation, 8-prompt eval (local 4-bit Instruct-8B)*. Top panel: mean KL (log scale) and perplexity vs the removed decoder layer; bottom: top-1 agreement and entropy. The shape is an **early cliff, not a symmetric U**: removing **L0** (KL ≈ 18 , top-1 $\approx 6\%$, PPL $\approx 1e4$) or **L1** (KL ≈ 14 , top-1 $\approx 23\%$) is catastrophic, while **L2–L29** sit at top-1 89–96% with KL ≈ 0.05 – 0.22 — individually removable — and only the final layer (L31: KL ≈ 0.98 , top-1 $\approx 88\%$) creeps up. Baseline entropy ≈ 0.21 nats, PPL ≈ 1.10 (teacher-forced on greedy output).

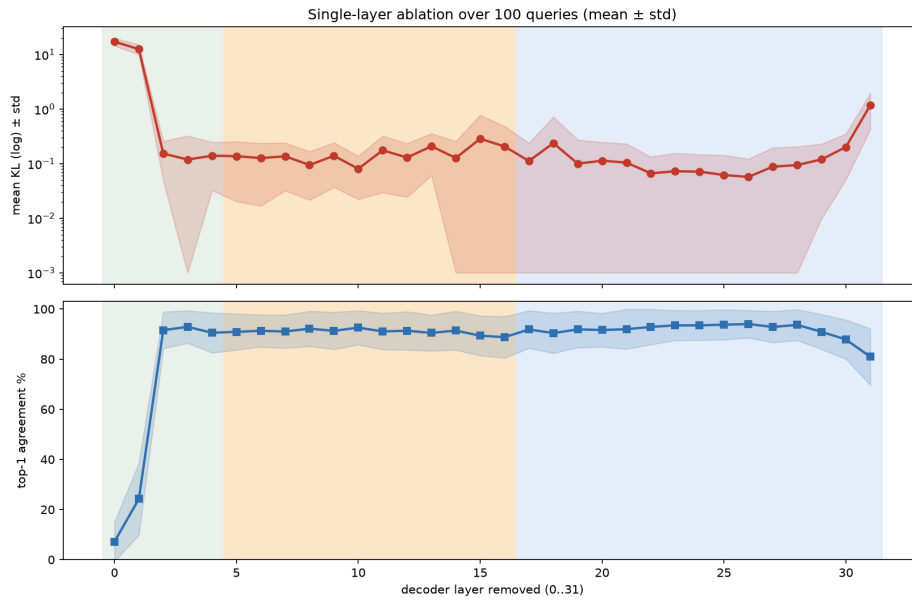


Figure 6

Figure 6 — the same sweep over 100 prompts, with mean $\pm$ std bands. The early cliff is confirmed with variance: only L0/L1 are catastrophic; **every layer L2–L29 sits at top-1 89–94%**. Robust, low-variance, and reproducible: exactly two layers are individually load-bearing.

5.3 Removing a band (Figures 5, 7)

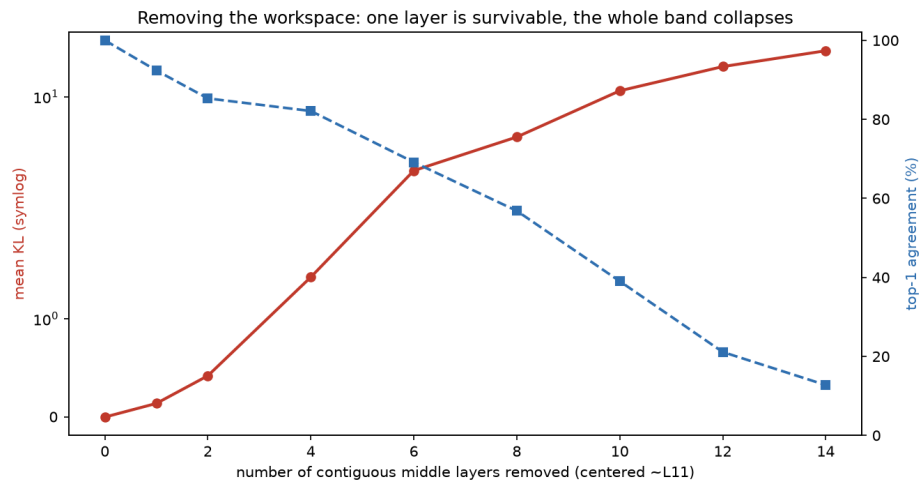


Figure 5

Figure 5 — cumulative removal, 8-prompt. As a growing contiguous middle band (centred ~L11) is deleted, KL rises and top-1 falls monotonically: 1 layer $\rightarrow$ 92%, then a graceful-then-catastrophic slide.

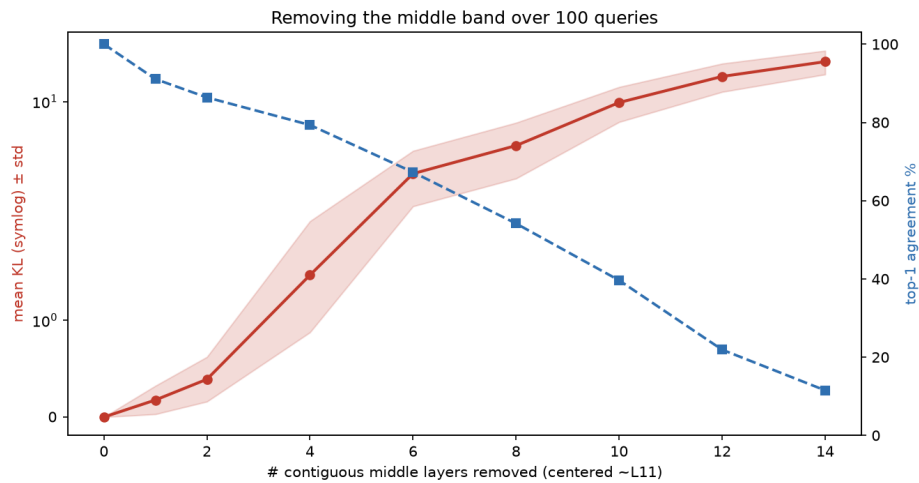


Figure 7

Figure 7 — cumulative removal, 100 prompts.

layers removed	1	2	4	6	8	10	12	14
KL	0.18	0.39	1.5	3.9	5.6	9.9	13.9	16.9
top-1	91%	86%	79%	67%	54%	40%	22%	12%

Interpretation (5.2–5.3): a single middle layer is redundant (the residual stream routes around it), but the middle band is *collectively* essential. Individually robust $\neq$ collectively expendable.

5.4 Which capability breaks first? (Figure 8)

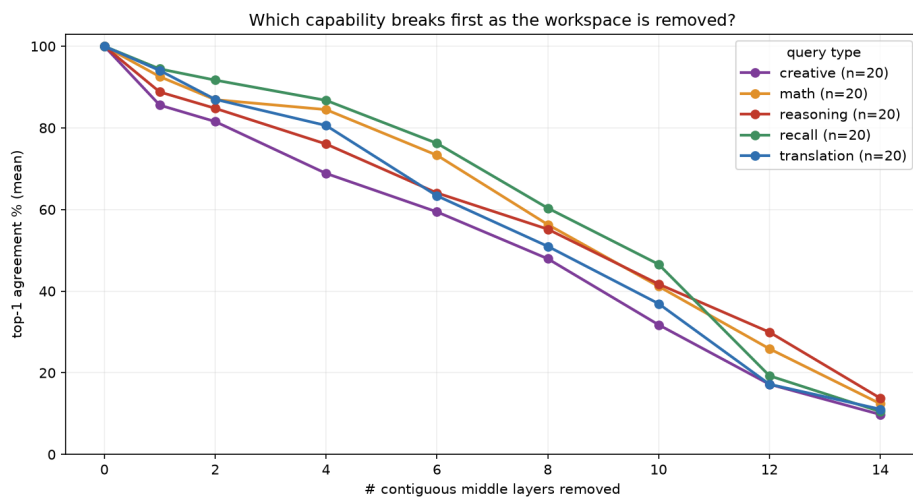


Figure 8

Figure 8 — band-removal effect split by capability (100 prompts, 20 each). Through the informative mid-range (4–10 layers removed) the ordering is consistent: **single-step recall is most robust and creative/abstract generation collapses first**, with math/reasoning/translation in between (e.g. at 6 layers removed: recall 76%, math 73%, reasoning 64%, translation 63%, creative 59%). Beyond ~12 layers everything converges toward collapse

and the ordering gets noisy. This is the **direction Anthropic’s J-space ablation predicts** — abstract, generative capability depends most on the middle band; shallow recall depends on it least. (Metric is top-1 agreement with the intact model’s greedy continuation — a proxy for “output preserved,” not a correctness grade.)

5.5 Full-precision replication via nnsight + NDIF (Figures 9–10)

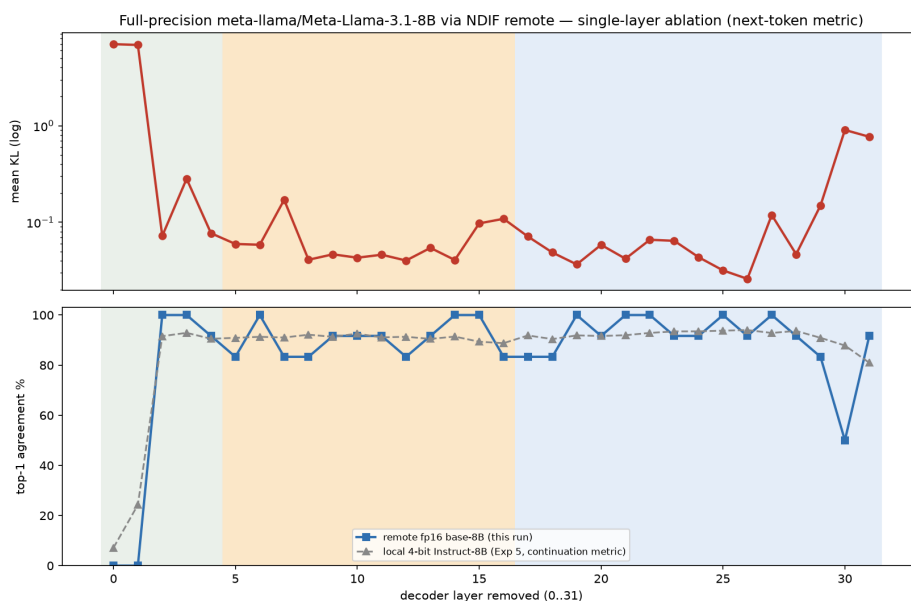


Figure 9

Figure 9 — single-layer sweep on full-precision base *Meta-Llama-3.1-8B* via NDIF remote (model executes on NDIF servers; the laptop only orchestrates — no local GPU). The **early cliff reproduces exactly** (L0/L1 → top-1 0%; L2–L29 → 83–100%), overlaid on the local 4-bit Instruct curve (grey), which it tracks closely across a change of **precision** (4-bit → fp16), **variant** (Instruct → base), **backend** (T4 → NDIF) and **metric** (continuation → next-token).

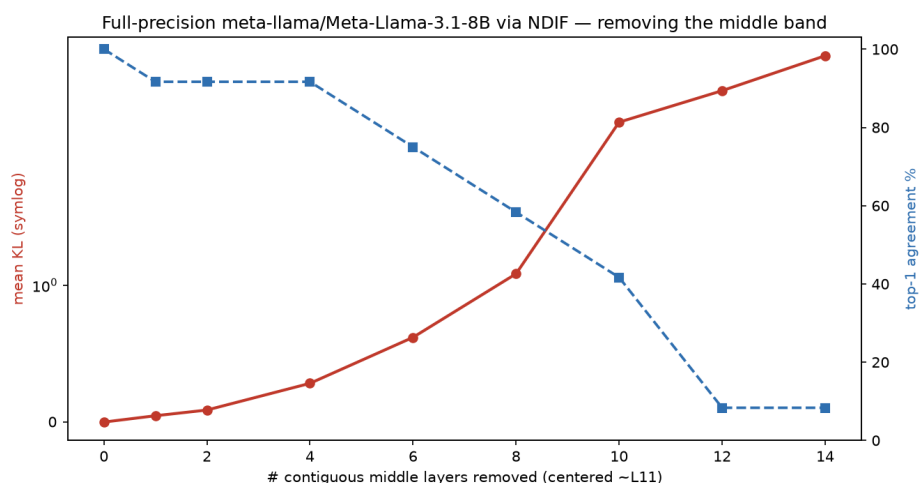


Figure 10

Figure 10 — cumulative band removal, full-precision 8B via NDIF. Same monotone collapse as the local runs. Conclusion: the “two load-bearing early layers + robust-but-collectively-essential middle” picture is a **property of the architecture, not of quantization or instruction-tuning**.

5.6 Scaling: 8B vs 70B vs 405B (Figure 11 + generation gallery)

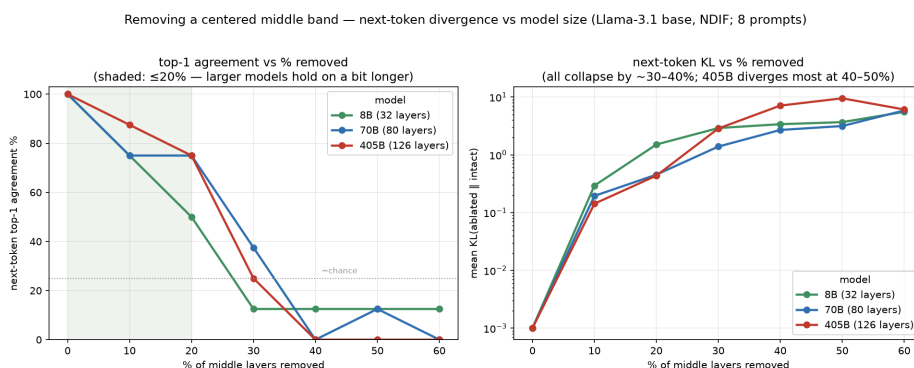


Figure 11

Figure 11 — removing a centred middle band of increasing size, all three base models via NDIF; next-token top-1 agreement (left) and KL (right) vs % of layers removed. Reading it honestly: at **low removal ($\leq 20\%$)** larger models hold on better — at 20% removed, **8B’s agreement has already halved (KL 1.5) while 70B and 405B are still $\sim 75\%$ (KL ~ 0.45)**. Beyond $\sim 30\%$ **all three collapse** on this strict metric, and at 40–50% the **405B distribution diverges the most** (KL $\sim 7\text{--}9.5$). So scale buys headroom at the *low-removal* end, but every size breaks by $\sim 30\text{--}40\%$ on the next-token metric.

Free-generation gallery (qualitative). The same prompt “*The theory of relativity states that...*” through 8B / 70B / 405B at 0 / 25 / 50% removal shows that **free generation is far more forgiving than the distributional metric**: all three stay grammatically fluent even at 50%, degrading into *artifacts* (repetition, stray multiple-choice, mild topic drift) rather than gibberish, e.g. 70B at 50% → “*The universe is made up of space and time. The universe is a vast expanse of space and time...*” (fluent, physics lost). The resolution of the apparent contradiction with Figure 11 is that KL/top-1 measure *divergence from the original model’s distribution*; a model whose predictions differ can still generate something coherent when left to sample freely. (Full transcripts accompany the repository.)

6. Discussion

A layer-level view of the workspace. Three findings line up with the global-workspace picture: (i) the middle band is where topic is *decodable but geometrically entangled* (Fig 3) — a low-variance, distributed code; (ii) that band is *collectively* necessary (Figs 5,7); and (iii) removing it harms *abstract/generative* capability before *shallow recall* (Fig 8) — the exact asymmetry Anthropic report when ablating J-space directions. Our instrument is a sledgehammer (whole layers) where theirs is a scalpel (low-rank directions), so this is a coarse, correlational corroboration, not a replication of the mechanism.

Redundancy and scale. The “early cliff” (only L0/L1 individually critical) and the greater tolerance of larger models at low removal (Fig 11) match the layer-pruning literature (Gromov et al.; ShortGPT): most middle layers are individually redundant, and redundancy grows with size. The twist is that at *high* removal the 405B distribution moves *most* — extra depth spreads the computation over more layers, so cutting half of them perturbs predictions strongly even as the model still produces fluent text.

Distribution shift $\neq$ incoherence. The single most important methodological caveat: KL and top-1 measure divergence from the intact model, and diverge much earlier than free-generation fluency degrades. Both are true and complementary; a study that reports only one will mislead.

Limitations. One model family; 4-bit locally; small eval sets (8/100 prompts); a top-1- agreement proxy rather than task accuracy; regime boundaries eyeballed from projections (the decodability curve, Fig 3, is the quantitative backbone); bands centred at the network middle (band *position* matters — the Instruct concept-band L6–L17 was more fragile than a centred band of equal size). Directional findings, not benchmarks.

Future work. Surgical *direction* ablation (project out top principal/probe directions rather than whole layers) to approximate J-space directly; the tuned lens for faithful per-layer decoding; task-accuracy grading (MMLU/GSM8K subsets) instead of top-1 agreement; and the *nnsight* recipes in the study appendix (logit lens, patching, steering, attribution) applied at 70B/405B scale.

7. Conclusion

Meaning in Llama-3.1 forms in a **middle band where information is decodable but entangled**. That band is individually redundant layer-by-layer but **collectively load-bearing**: removing it degrades the model

gracefully-then-catastrophically, hitting **abstract/creative** capability before **factual recall**. The pattern holds across quantization, instruction-tuning, backend and metric, and across model sizes — with larger models more tolerant at low removal. This is a blunt, reproducible, single-GPU-plus-remote corroboration of the “verbalizable global workspace” view of intermediate representations.

8. Glossary

- **Residual stream** — the additive running hidden state each layer reads from and writes to; the model’s internal “bus.”
 - **Ablation** — replacing a component with identity/zero/mean to test its necessity.
 - **Discard-delta / rebind** — two implementations of identity-ablating a layer (run-and-discard the update; or set the layer’s output to the previous layer’s output).
 - **Entropy $H(\mathbf{p})$** — uncertainty of a single distribution, $-\sum p \log p$.
 - **Cross-entropy $H(\mathbf{p}, \mathbf{q})$** — $-\sum p \log q$; PPL = $\exp(H)$.
 - **KL divergence $D_{\text{KL}}(\mathbf{p} \parallel \mathbf{q})$** — relative entropy
 $\sum p \log(p/q) = H(\mathbf{p}, \mathbf{q}) - H(\mathbf{p})$; ≥ 0 , asymmetric.
 - **Perplexity** — $\exp(\text{cross-entropy})$; lower = the model is less “surprised.”
 - **Top-1 agreement** — fraction of positions where the ablated and intact models pick the same most-likely next token.
 - **Decodability (k-NN accuracy)** — how well a simple classifier reads the label off a layer’s activations (information *presence*).
 - **Silhouette score** — $(b-a)/\max(a, b)$; how visually separated clusters are in a projection (information *geometry*).
 - **UMAP / t-SNE** — nonlinear 2-D projections for visualization (not distance-faithful).
 - **Logit lens / tuned lens** — decoding an intermediate residual through the unembedding to read the model’s “current guess.”
 - **Activation patching / causal tracing** — copying clean activations into a corrupted run to localise causal information.
 - **Steering vector** — a direction added to activations to move behaviour along a concept.
 - **Superposition** — many features packed into overlapping directions (why entangled $\neq$ absent).
 - **J-space / global workspace** — Anthropic 2026: a small set of intermediate token-aligned directions that behave like a shared reasoning “blackboard.”
 - **NDIF / nnsight** — remote execution of, and intervention on, foundation-model internals; lets a laptop run experiments on 70B/405B.
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Reproducibility: all figures are generated by scripts in experiments/; metrics and per-query outputs are in blog/figures/.json and *.csv; the interactive server/UI and the nnsight recipes accompany the repository.*